

deSitter zero-representation and YM inspirations

Maximilian Weinhold

May 2026

1 Introduction

Let us create a pair of Turing machines:

qu, lo

Given the number of rules present when the Turing machines run an identical program $\mathcal{M}$, they will lead to different patterns: $\mathbf{qu}(\mathcal{M}) := 0 \rightarrow 0 \rightarrow 0 \rightarrow 0 \rightarrow 0 \rightarrow \dots$ and $\mathbf{lo}(\mathcal{M}) := 0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0 \dots$

With Occam's razor in mind, we say that **qu** is a one-step elif: $if \mathcal{M}(0); 0 \text{ else } \mathcal{M}(j); 0$ allowing us to absorb the first entry and create a subsequently always true, and similarly say **jo** is a two-step exclusion: $if \mathcal{M}(j); 0 \otimes if \mathcal{M}(0); j$ and we let $j = 1$

So that we may say $Kol(\mathbf{jo}(\mathcal{M})) \geq 2$ and $Kol(\mathbf{qu}(\mathcal{M})) \geq 1$ where we envision a Kolmogorov complexity requiring at least a 2-state descriptor for **jo**. If we hide the first entry, $Kol(\mathbf{qu}(\mathcal{M})) = 1$

Associating **qu** with the process that flows the Yang-Mills action along a K-algebra provided with a suspension of the Hubble parameter $t \cdot RdS$ to create¹ the object: $m(0) \rightarrow m((t \cdot R)dS) \rightarrow m(0)$ where m recovers the path of a particular realization which I want to say is strung as a particular value along a Bott periodicity, and with I think the last association

$$m(0 \rightarrow (t \cdot R)dS \rightarrow 0) : \mathbf{jo} \quad ; \quad m(0) : \mathbf{qu}$$

and assuming we have settled past the first entry, we may say that

$$Kol(\mathbf{jo}_{YM}^{susp}) - 1 > Kol(\mathbf{qu}_{m=0}) - 1 = 0$$

Which resolves as a mass-gap with a notion of mass as the Kolmogorov complexity of a described Turing process; a 'heavier' program.

¹Sorry, I think this is not precise. I will be researching the support form: $S_{YM}^{susp} = \int_{\partial dS \times [0,1]} d^4(t, \theta) e^{2Ht} Tr(F \wedge *F)$